

ImageNet - dataset  
images about daily obj & animals.  
↓  
24 million of 2k classes

Date: \_\_\_\_\_

Day: \_\_\_\_\_

## Transfer learning:-

T.L is a technique that solve of problem data which is not present in pretrained models.

- T.L is a research problem in ml that focuses on storing knowledge gained while solving 1 problem & applying it to a diff but related problem.

e.g:- riding a bicycle & shifting to bike.

How:-

replace VGGNet with 2

[Conv. base] [□ □ □] neuron → binary.  
sigmoid

freeze

F.c layer.

→ So def new b/m:-

Using already learned things by a model using that & applying it on your own task.



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Feature extraction

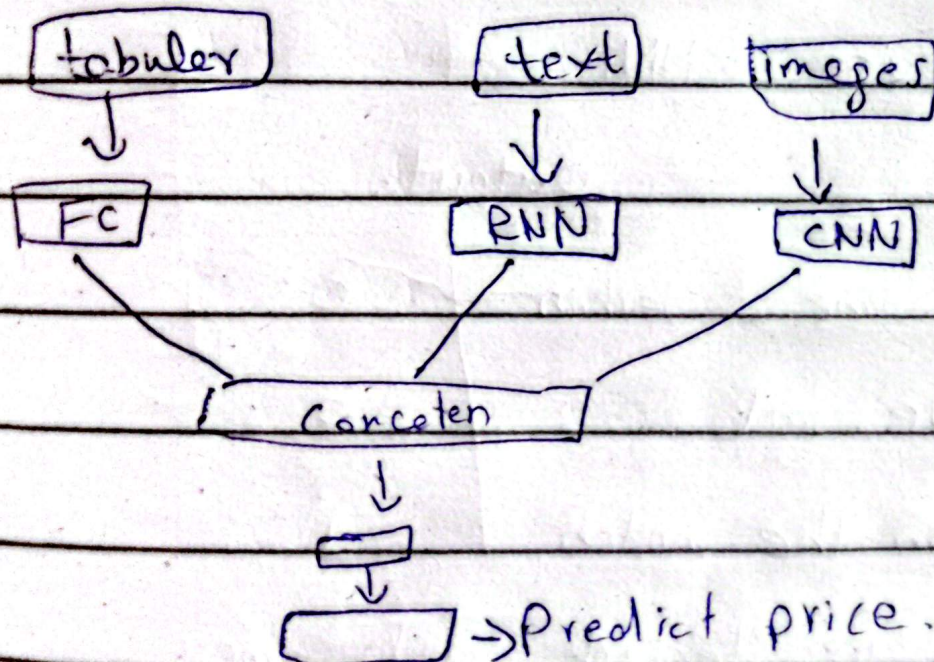
adding our own  
just dense layers  
at last

Fine tuning.

Adding dense layer  
along with some conv2D  
layers also.

Function models:-

It is building non-linear  
neural networks.



→ Non-linear topology → func API  
in keras.

→ multiple inputs & outputs.